PC 503- Programming Lab

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Course Placement: Programming Lab is a PG-Program core course for 1st year students of M.Tech. ICT (SS, ML, and VLSI & ES specializations) and M.Tech. (EC) programs; Autumn, 2025-2026.

Course Format: Credit Structure (L-T-P-Cr): 1-0-4-3; 1-hour lecture and 4 hours' lab every week.

Course Content: This course aims to provide hands-on practice in software tools and technologies to M.Tech (ICT) and M.Tech (EC) students. The broad coverage of this course is as follows: Familiarity in Linux; Shell Programming; Programming tools (Make files, version control, debugger, GitHub). Problem solving and programming using Python, Introduction to Circuit Modelling and Analysis; System Design, Lab on Sampling and Quantization.

A single textbook is insufficient for this course. Some of the relevant resources that will be referred to are:

Reference Book:

1. Introducing Python: Modern Computing in Simple Packages by Bill Lubanovic, O'Reilly, 2nd Edition, 2019.
2. Coding Python and SQL: How to Install and Python Code: How to Improve SQL and Python by Patricia Jasmine, 1st Edition, 2022.
3. Learning the Pandas Library: Python Tools for Data Munging, Analysis, and Visualization by Matt Harrison and Michael Prentiss, 2nd Edition2016.
4. Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Ipython by Wes McKinney, Second Edition, 2017.
5. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems by Aurelian Gerona, 2nd Edition, 2019.
6. Flask Web Development: Developing Web Applications with Python by Miguel Grinberg, 2nd Edition, 2018.
7. Django3 Web Development Cookbook: Actionable solutions to common problems in Python web development by Aidas Bendoraitis and Jake Kronika, 4th Edition, 2020.
8. Web Scraping with Python: Collecting More Data from the Modern Web by Ryan Mitchell, 1st Edition, 2018.
9. Python GUI Programming - A Complete Reference Guide: Develop responsive and powerful GUI applications with PyQt and Tkinter by Alan D. Moore and B.M. Harwani, 1st Edition, 2019.

10. Python for MATLAB Development by Albert Danial, 1st Edition, Apress Berkeley, CA, 2022.
11. Arduino Cookbook: Recipes to Begin, Expand, and Enhance Your Projects by Michael Margolis, 2nd Edition, 2021.
12. Exploring Arduino: Tools and Techniques for Engineering Wizardry by Jeremy Blum, 2nd Edition, 2019.
13. The Elements of MATLAB Style by Richard K. Johnson, 1st Edition, Cambridge University Press, 2010.

Related Resources:

1. Visual Studio Code: https://code.visualstudio.com/download
2. Anaconda: https://www.anaconda.com/download
3. Python 3.9: https://www.python.org/downloads/
4. Tinkercad, Autodesk: https://www.tinkercad.com/dashboard
5. LTspice:https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html
6. MATLAB (licensed version available at DA-IICT): https://in.mathworks.com/campaigns/products/trials.html

The Project Topics (Not Limited):

- Hand Gesture Recognition
- Emotion Recognition
- Signal Denoising
- Embedded AI and ML
- Voice Recognition
- Image Processing
- Gait Analysis
- Kidney Disease Diagnosis
- Face Recognition
- Biomedical Signal Processing
- Robotics
- Real time weather forecasting

Assessment Method: Project and Assignments, a mid-semester and a final examination. Evaluation Scheme: i) Hands on Lab and Attendance: 20% ii) End-sem Lab Exam: 30% iii) Project and Assignments: 50%.

Course Outcomes: This course will give an opportunity to the MTech-ICT (SS, ML, VLSI&ES) and MTech (EC) students to gain skills in Python, MATLAB, familiarity in programming and EDA tools, Database and Web Development, Data Science, AI/ML, Web Scraping and GUI development. The course consists of two parts which are (i) interactive lecture section and (ii) hand-on which will provide a platform to learn programming from scratch to expert level with many hundreds of examples, and more than 100 exercises. Students will learn Pandas, NumPy, and Matplotlib for DS (Data Science), Scikit-learn for ML (Machine

Learning), Soup, Scrapy Spiders and tkinter for GUI (Graphical User Interfaces). Students will acquire skills to deploy the code in prototype systems to solve real-world problems. The course is also focus on a wide variety of projects such as data visualization, modelling, spectral analysis, data clustering, image processing, filtering, embedded-AI/ML, scientific computing which will be performed by the students individually or in a group. Today, world leading companies are looking for the people who are expertise in programming, used in the area of data science, AI/ML, web development, web-scraping, GUI development, and edge-computing. This course will provide a platform to develop skills for which companies are interested to hire professionals.

Mapping:

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
X	X			X	X			X	X	X		X	X	

Lecture Schedule (Includes Lectures and Labs Hours)

S. No	Description		No. of Lectures
1	**Introduction to Subject**, Syllabus, Text and Reference Books Discussion, Objectives, scope, outcomes of the course, and evaluation scheme		**01**
2	**Introduction to Python**		**08**
	2.1	Introduction- Installation of necessary software- different IDE platforms	01
	2.2	Variables- Datatypes-Logic-Control Flow	03
	2.3	Loop – Functions	02
	2.4	List-Dictionaries-Tuples-Sets	02
3	**Advanced Python**		**08**
	3.1	Error Handling- Exceptions	02
	3.2	Object Oriented Programming	02
	3.3	Iteration- Generators-Closure-Decorators	02
	3.4	Modules- Standard Library- Pypi	02
4	**Data Analysis for Data Science**		**10**
	4.1	Data Analysis- Data Manipulation- Pandas	02
	4.2	Data Computation- Data Analysis- Numpy	02
	4.3	Data Visualization- Matplotlib	02
	4.4	Advanced Plotting- Seaborn	02
	4.5	Interactive Visualization	02

5	**Machine Learning (ML) and Deep Learning (DL) Implementation**		**12**
	5.1	Introduction to Machine Learning and Deep Learning	01
	5.2	ML/DL Workflow for Deployment in DSP and Embedded Systems	01
	5.3	Implementation of simple Machine Learning Models with Scikit-learn library	04
	5.4	Extraction of features from the dataset	02
	5.5	Implementation of Simple Deep Learning Models (CNN, LSTM, GRU) using Tensor Flow	04
6	**Web Scrapping Essentials**		**04**
	6.1	Basic of Web Scrapping and Beautiful Soup Parse Tree	02
	6.2	Scrapy Shell – Scrapy Spiders	02
7	**Graphical User Interface**		**04**
	7.1	Getting started with GUI	01
	7.2	Development of GUI App with Tkinter	02
	7.3	GUI for Prototype System	01
8	**Arduino, Circuit Simulation, and EDA Tool Integration**		**12**
	8.1	Introduction to Embedded System & Arduino Microcontrollers	02
	8.2	Basic Circuit Designing using EDA Tools (Tinkercad circuits/ Proteus)	04
	8.3	Arduino Programming: Digital I/O, Sensors, and Actuators	04
	8.4	Arduino-Python Integration: Real Time Data Processing	02

*The extent of topic coverage will be influenced by the collective engagement and participation of the class.